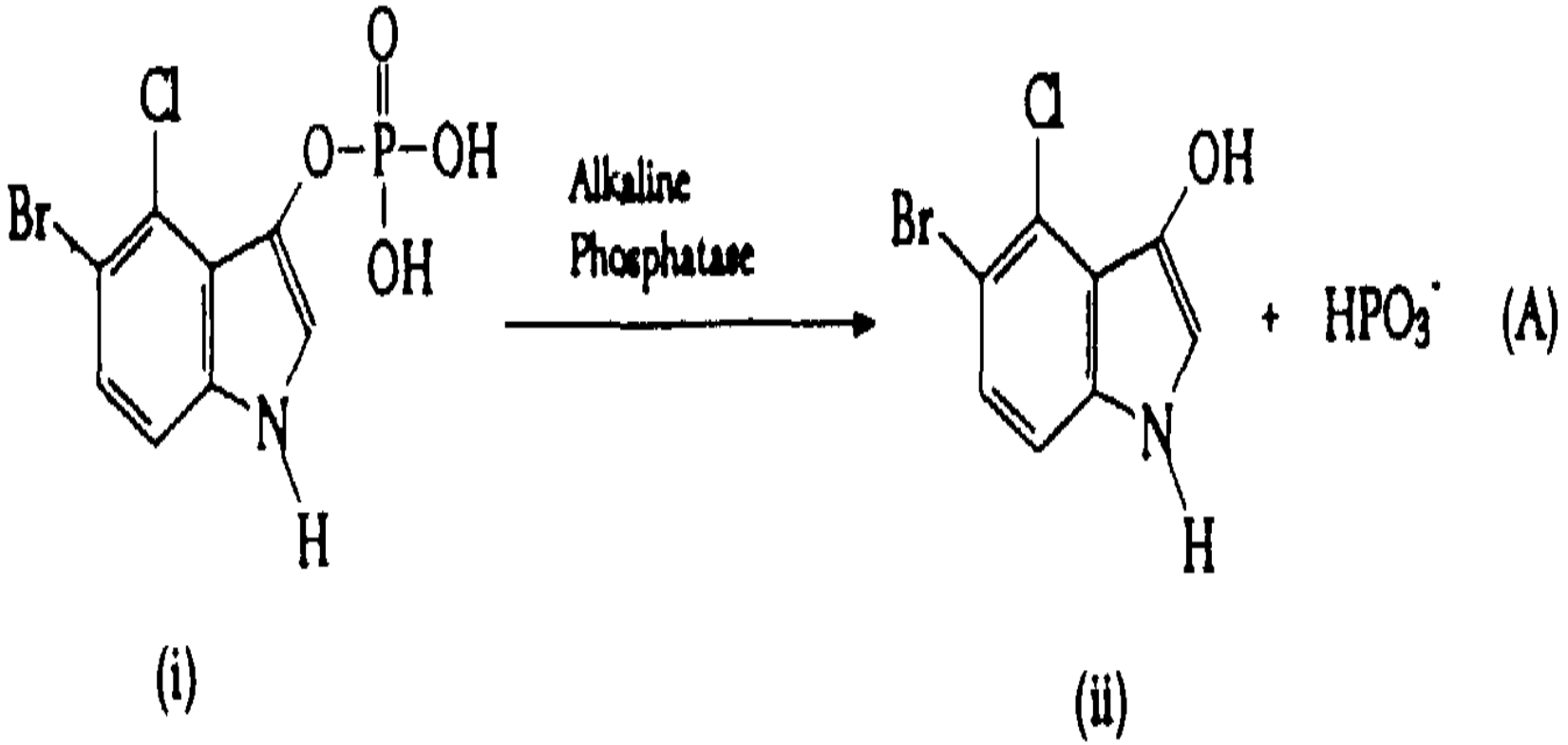
INTERPRETATION OF ALKALINE PHOSPHATASE & AMINOTRNASFERASES

INTRODUCTION

- Hepatic enzymes that are usually intracellular, but are released from hepatocytes (with hepatocellular injury).
- In liver diseases - serum levels of various cytosolic, mitochondrial and membrane associated enzymes are increased.
- Alanine aminotransferase (ALT), Aspartate aminotransferase (AST), Alkaline phosphatse (ALP) and Gamma glutamyl transferase (GGT) are the enzymes most often measured for evaluation of liver disease.
- Distinguish among different types of liver disorders.
- Gauge the extent of known liver damage

Alkaline Phosphatase (ALP)

- **Hydrolase** enzyme (removal of Phosphate group).
- A group of enzymes that catalyze the hydrolysis of a large number of organic phosphate esters.
- Most effective in alkaline environment.
- ALP – membrane bound glycoprotein enzyme.


Cl
O
O–P–OH
OH
Br
N
H
Alkaline
Phosphatase
Cl
OH
Br
N
H
+ HPO3•
(A)
(i)
(ii)

- Present in highest concentration in the sinusoids and in the endothelium of the central and peri-portal veins.
- Smaller concentrations occur in the biliary canaliculi.
- In liver, - active role in down-regulating the secretory activities of the intrahepatic biliary epithelium.
- Clearance of liver enzymes from plasma - average t½ life of ALP is 9.7 days.

- Present in –
  - Liver
  - Bone
  - Intestine
  - Placenta (first trimester)
  - Kidney
  - Pluripotent stem cells have elevated levels of alkaline phosphatase on their cell membrane.
- Estimation in serum - ALP catalyzes the hydrolysis of ρ-Nitrophenyl phosphate (pNPP) to ρ-Nitrophenol

Enzymes - Serum Levels

- ***Normal Serum levels*** - indicates balance between its synthesis and release (routine cell turnover).
- ***Raised Enzyme levels*** – due to;
  - Cellular damage
  - Increased (rate) cell turnover
  - Proliferation of cells
  - Increased synthesis
- **Markers to detect the cellular damage .**
- **In the serum, two types of ALP isozymes predominate: skeletal and liver.**

ALP – Reference Ranges

Adults	25-100 U/L
Children	Less than 350 U/L

ALP – Diagnostic Marker

- ***Serum ALP*** - biochemical marker to investigate ***hepatobiliary disease*** and ***bone disease*** (associated with increased osteoblastic activity).
- The increased activity of ALP in diseased human liver is believed to be due to enhanced synthesis.
- Elevated ALP – occurs with all forms of cholestasis

- Serum of Cholestasis patients - fragments of hepatocyte membrane rich in ALP (and also GGT) activity have been detected.
- It may due to the release of cell membrane fragments into circulation.
- In cholestatic disease, the stasis and prolonged contact of bile acids, with canalicular and bile duct epithelia could solubilize and release ALP (and also GGT) from plasma membrane.

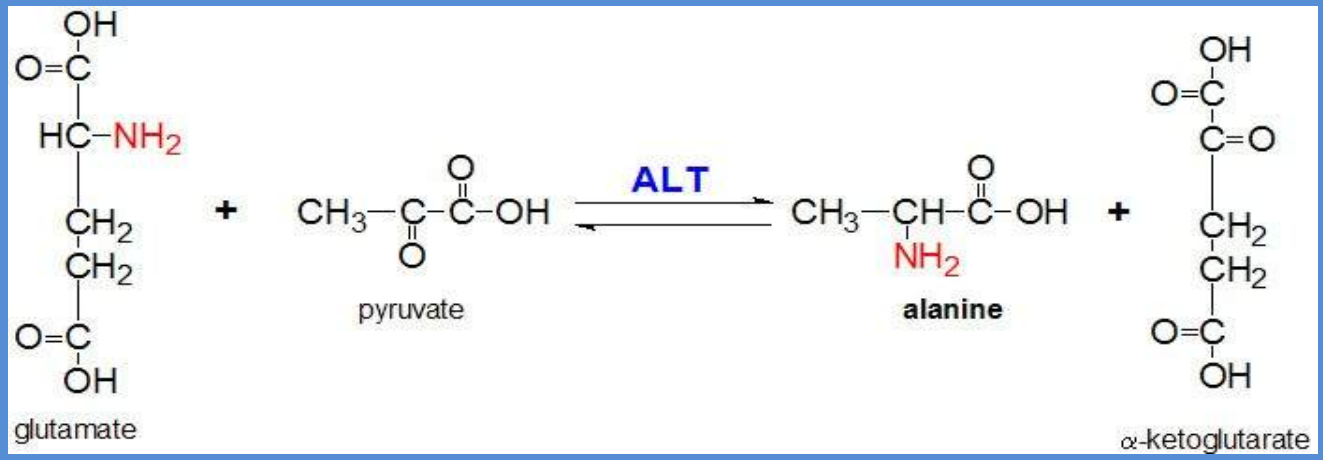
- Serum ALP levels - are dependent on age factor and its elevated levels are also observed during periods of active bone growth.
- High serum ALP is also accompanied by increased osteoblastic activity.
- Paget's disease (osteitis deformans), osteoblastic tumors with metastases.
- Hyperparathyroidism when there is mobilization of Ca and P from the bone, rickets and osteomalacia.
- High ALP - Heart failure, heart attack, mononucleosis, or kidney cancer.

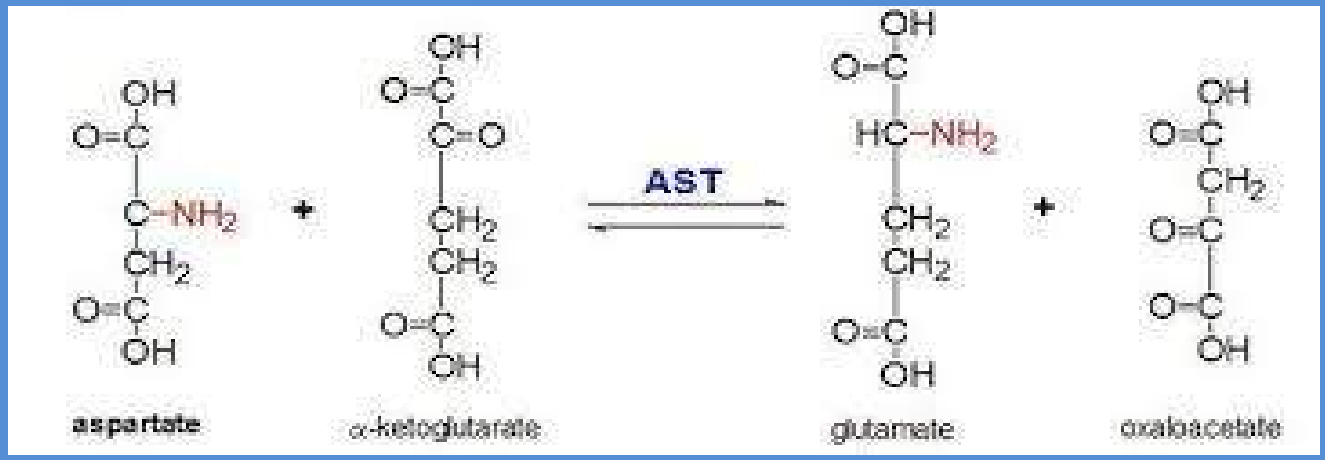
- High ALP - any serious infection that has spread through the body (sepsis).
- High ALP - In pregnancy (3rd trimester), because the placenta makes ALP.
- Low ALP levels - malnutrition conditions such as celiac disease or are caused by a lack of nutrients in the diet (such as scurvy) can cause low ALP levels.

AMINOTRANSFERASES

- Enzyme that catalyzes a type of reaction between an amino acid and an α-keto acid.
- Actively involved in the synthesis of amino acids, which form proteins.
- Importantly, they are key indicator of liver damage.
- Two important aminotransferase enzymes are;

1. ***Aspartate aminotransferase*** (AST) OR serum glutamic oxaloacetic transaminase (SGOT)
2. ***Alanine aminotransferase*** (ALT) OR serum glutamate-pyruvate transaminase (SGPT).

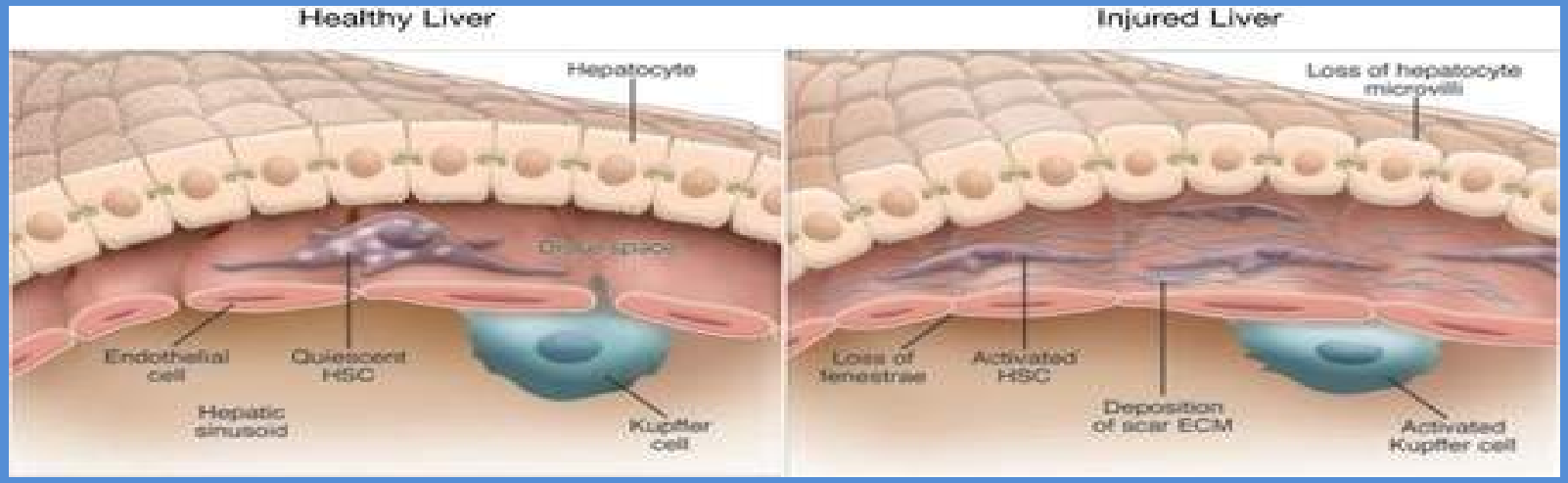

ALT
pyruvate
alanine
glutamate
α-ketoglutarate

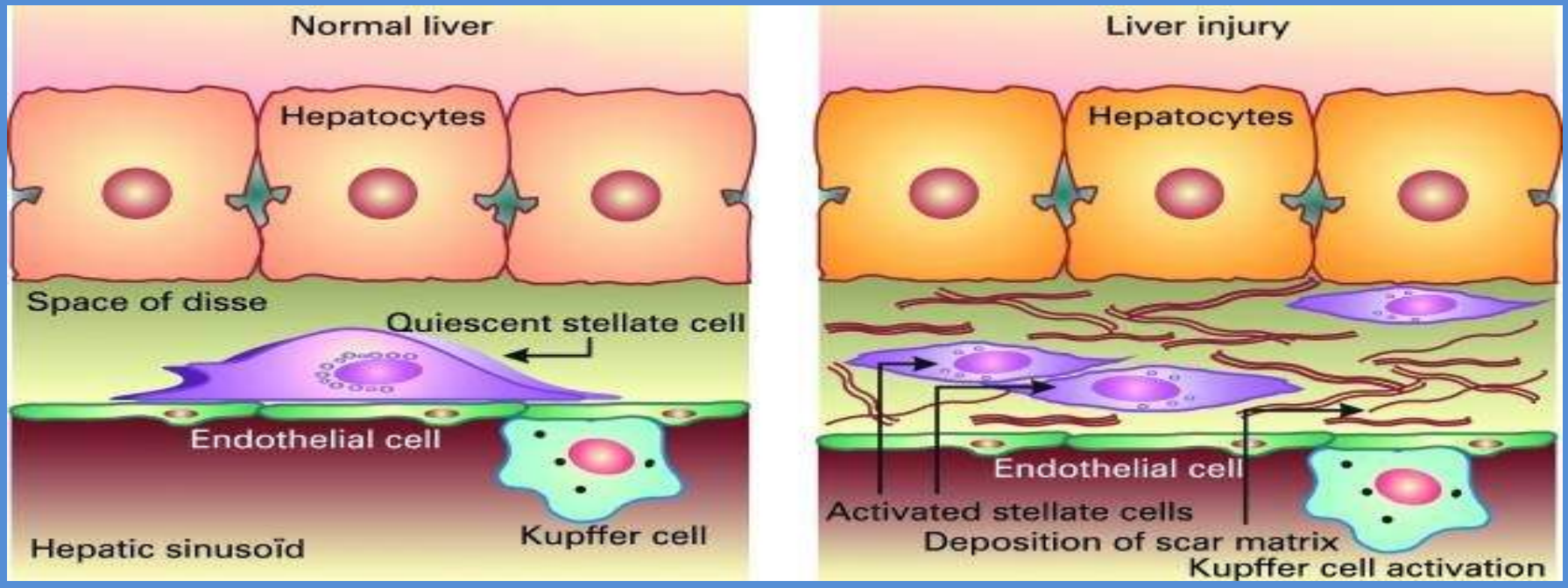

AST
aspartate
α-ketoglutarate
glutamate
oxaloacetate

- AST – occurs in hepatocytes (in mitochondrial and cytosolic isenzyme forms).
- ALT – is also present both in mitochondria and cytosol.
- The mitochondrial is low in activity and nearly absent in serum, even in disease.
- Key factors that affect the degree of elevation of these enzymes;

1. Specific enzyme activities in hepatoctyes.
2. Relative rate of catabolic clearance of these enzymes from the bloodstream.

- In chronic liver disease – the activity of ALT (in liver tissue) decrease to a significant greater degree than does that of AST.
- In acute hepatitis – activities of ALT & AST decreases significantly.
- In liver cirrhosis – higher serum activity of AST than ALT.
- ALT is high – acute hepatitis (than AST).





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- In hepatic damage – permeability of the hepatocyte membrane increases.
- Cytosolic isoenyzme of aminotransferases spills into the sinusoids and then into the peripheral blood.
- Permeability of the mitochondrial membrane also increases and mitochondrial isoenzymes are then released as well.
- Clearance of liver enzymes from plasma - average t½ life of ALT is 47 hours and AST (cytosolic) is 17 hours.

ALT & AST – Reference Ranges

ALT	3-40 IU/L
AST	4-45 IU/L

INTERPRETATION

- Certain patterns exist with LFTs
  - **Hepatocellular Injury**: Very high AST, ALT with mild/moderately elevated alkaline phosphatase.
  - **Cholestatis**: mild/moderately elevated AST/ALT with very high alkaline phosphatase
- **Alcohol** Use:
  - Frequently have AST:ALT ratio ≥ 2:1
  - History: Need accurate assessment of alcohol intake, etc

AST/ALT ratio

- Normal is 0.8
- In alcoholic hepatitis, is usually > 2

- Medications:
  - History: Need to assess temporal relationship with drug, see if patient improves once medication removed
  - **NSAIDs, antibiotics, statins, anti-tuberculosis medications, anti-epileptic drugs, acetaminophen**
    - Frequently cause isolated elevated aminotransferases
  - **Acetaminophen overdose**
    - Toxicity is likely to occur with single ingestions greater than 250 mg/kg or those greater than 12 g over a 24-hour period
    - AST/ALT elevations is first sign of liver damage (usually 24-hours after ingestion)